Transformers

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Agenda

- Mecanismos de atención
- Arquitectura de la red neuronal tranformer
- BERT (Bidirectional Encoder Representations From Transformers)
- Transferencia del aprendizaje
- Caso de uso real: Clasificación de registros poblacionales de cáncer

Español:

- Es hora del té

Ingles:

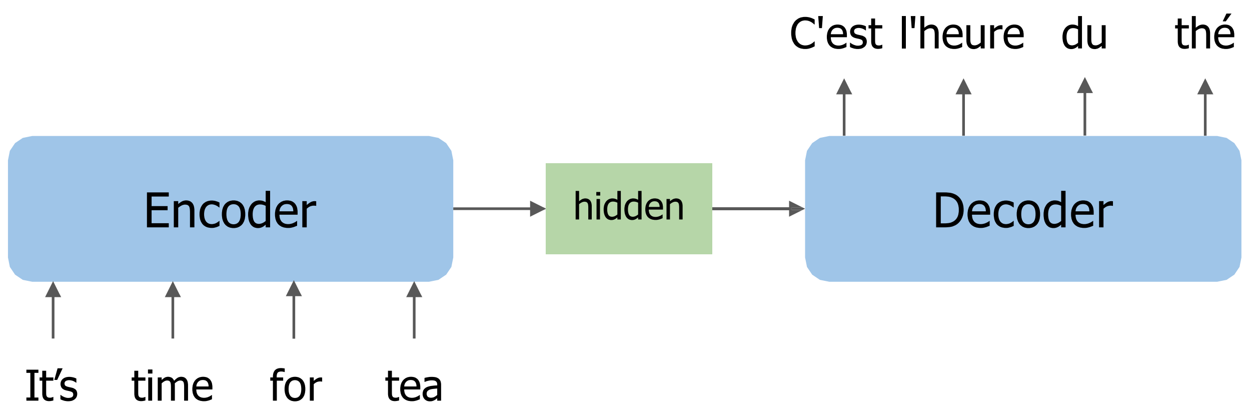
- It´s time for tea

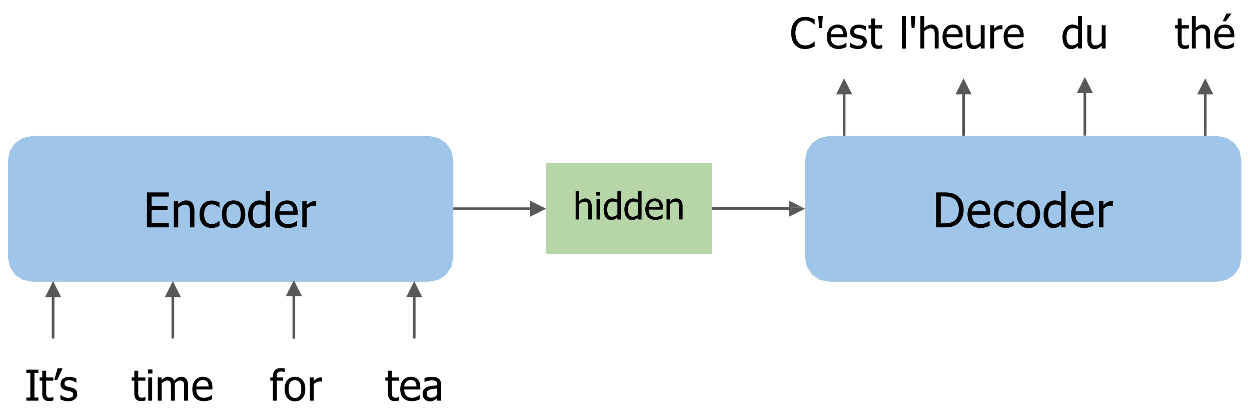
Ingles

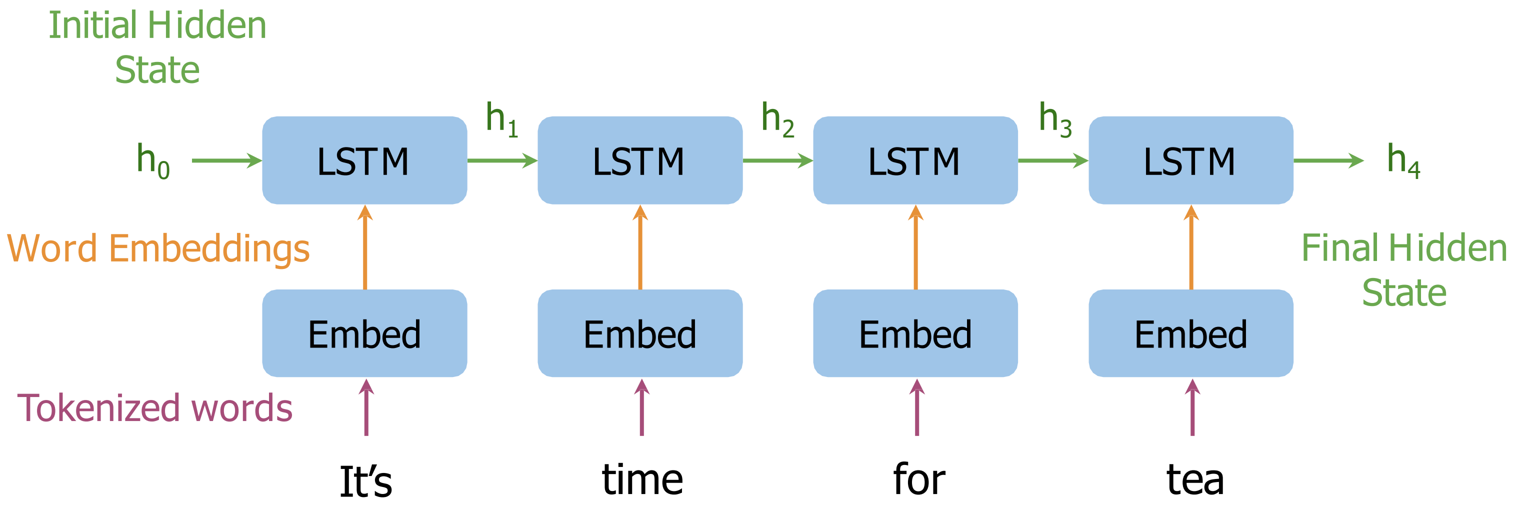
- It´s time for tea

Frances:

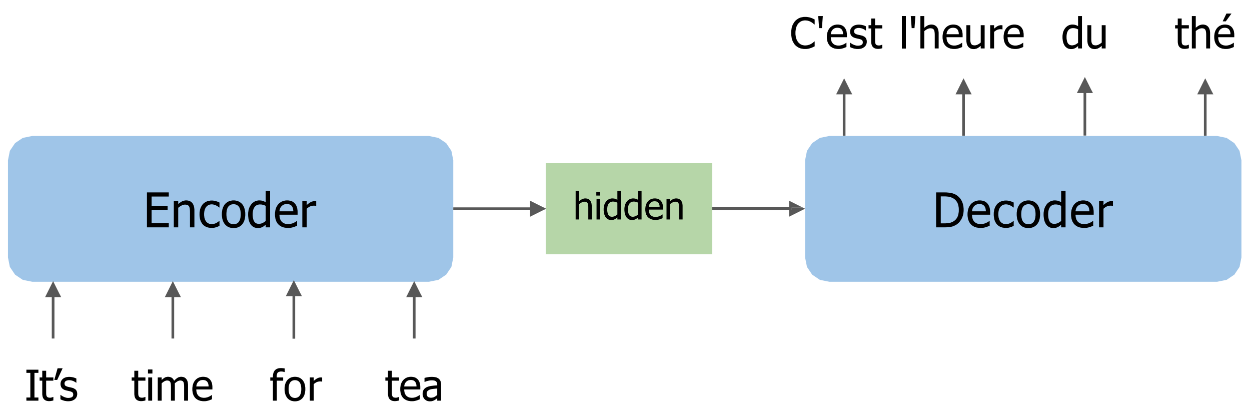
- C'est l'heure du thé

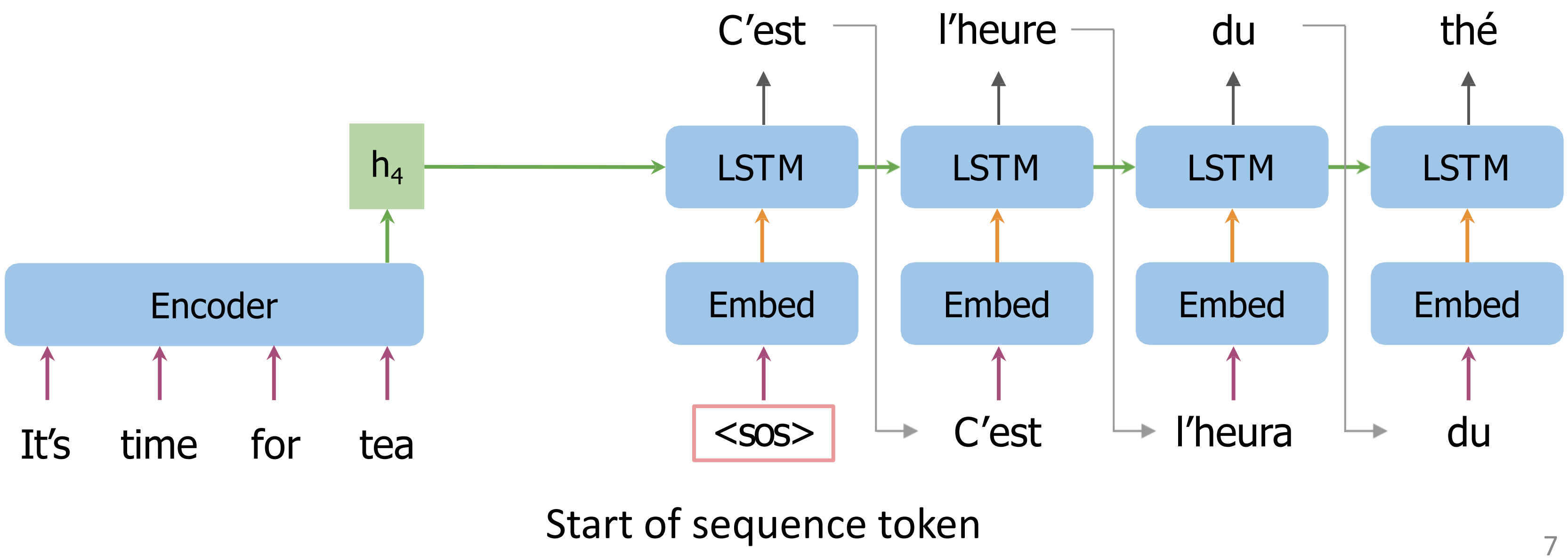

C'est l'heure du thé
Encoder
hidden
Decoder
It's time for tea

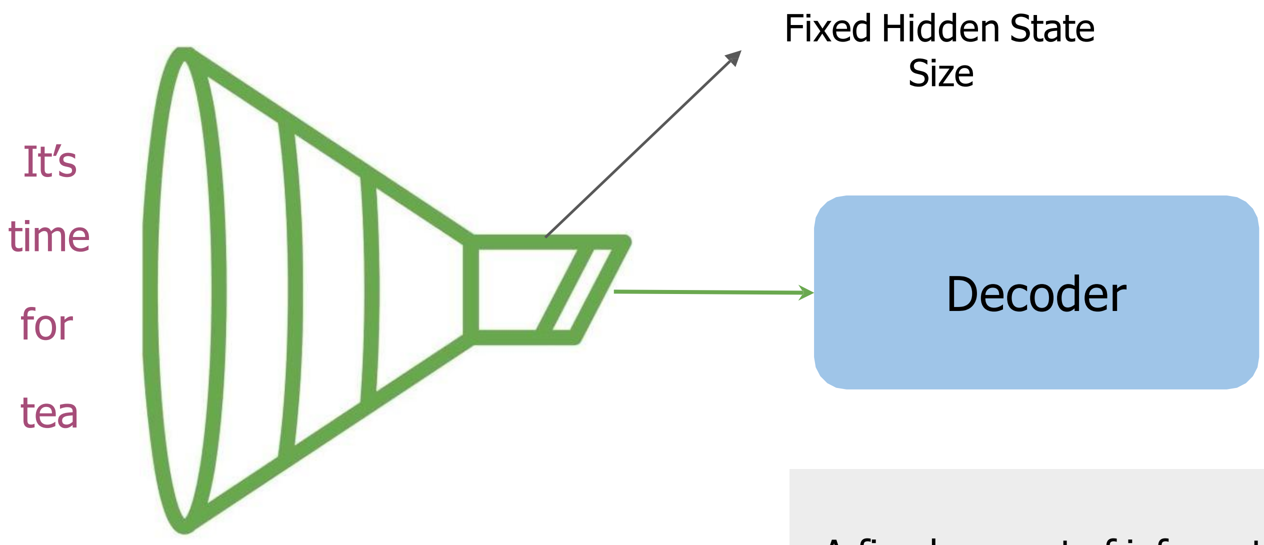




Encodes the overall meaning of the sentence

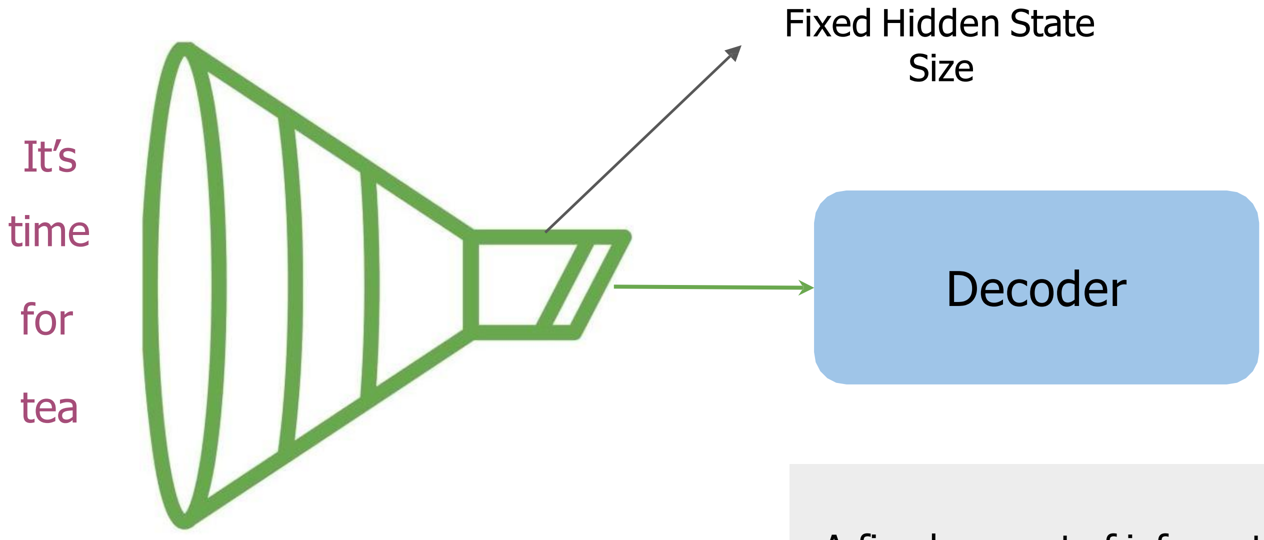

C'est
l'heure
du
thé
Encoder
hidden
Decoder
It’s
time
for
tea


C’est
l’heure
du
thé
h_4
LSTM
LSTM
LSTM
LSTM
Encoder
Embed
Embed
Embed
Embed
It’s
time
for
tea
<sos>
C’est
l’heura
du
Start of sequence token

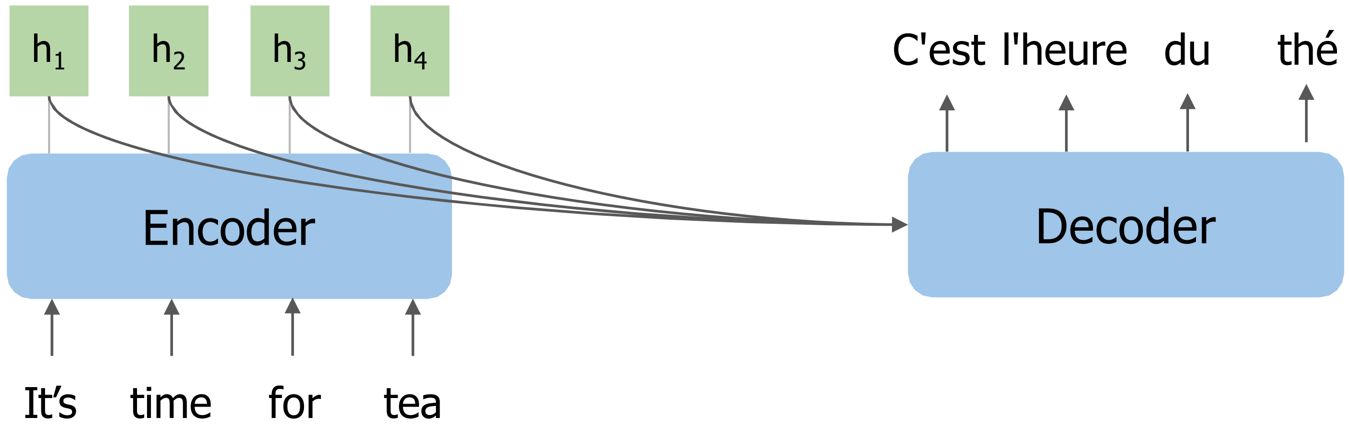


A fixed amount of informations goes from the encoder to the decoder

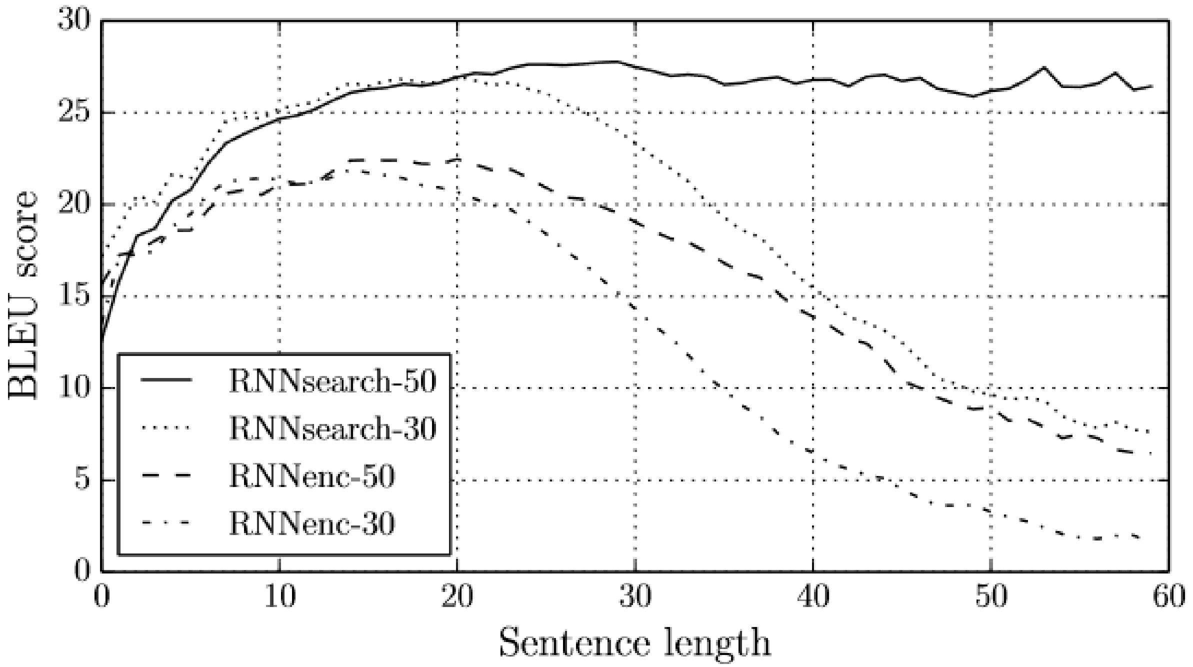
Variable-length sentences to fixed-length memory = As sequence size increases, model performance decreases

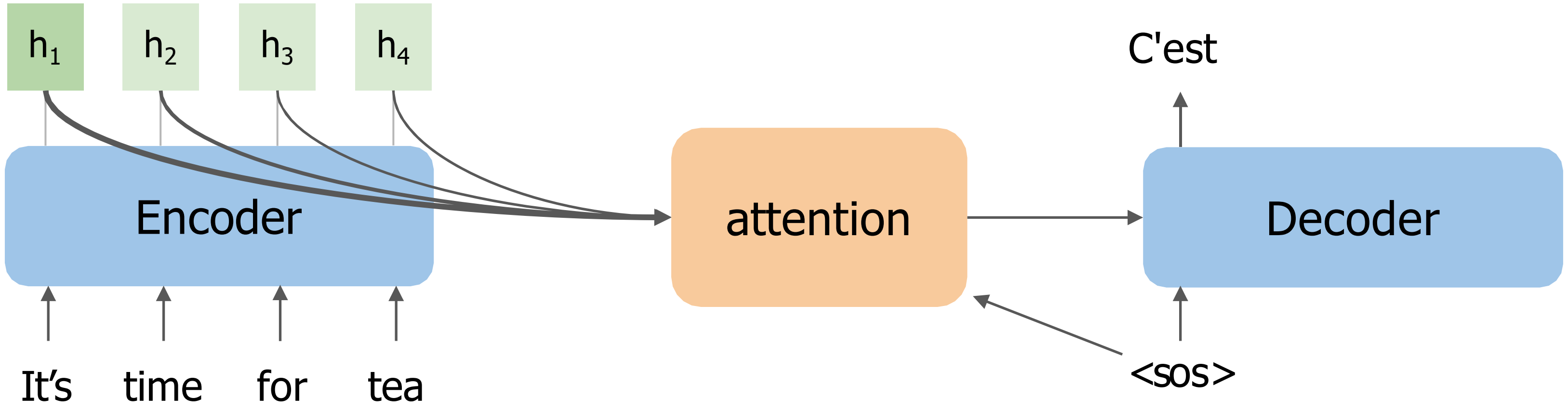


A fixed amount of informations goes from the encoder to the decoder

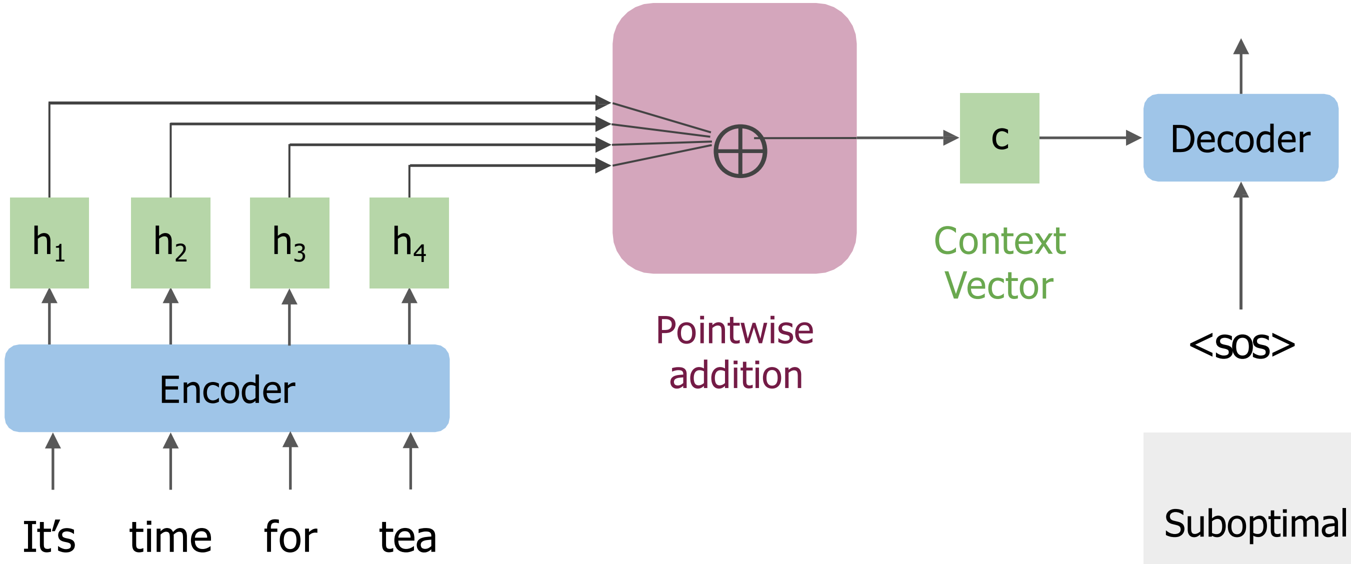


Bahdanau, D., Cho, K., & Bengio, Y. (2016). *Neural Machine Translation by Jointly Learning to Align and Translate* (arXiv:1409.0473). arXiv. https://doi.org/10.48550/arXiv.1409.0473

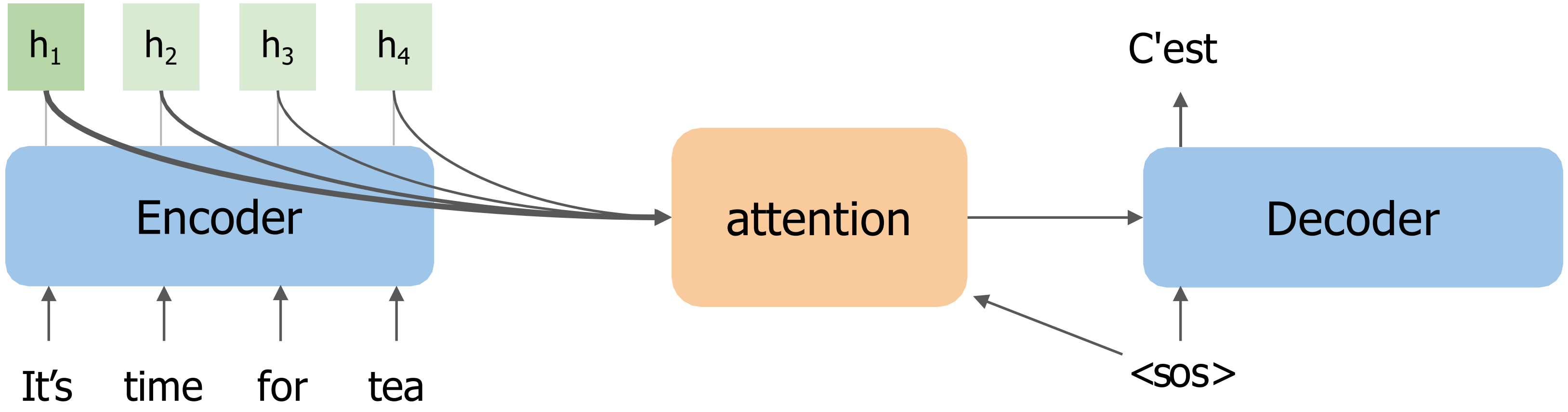




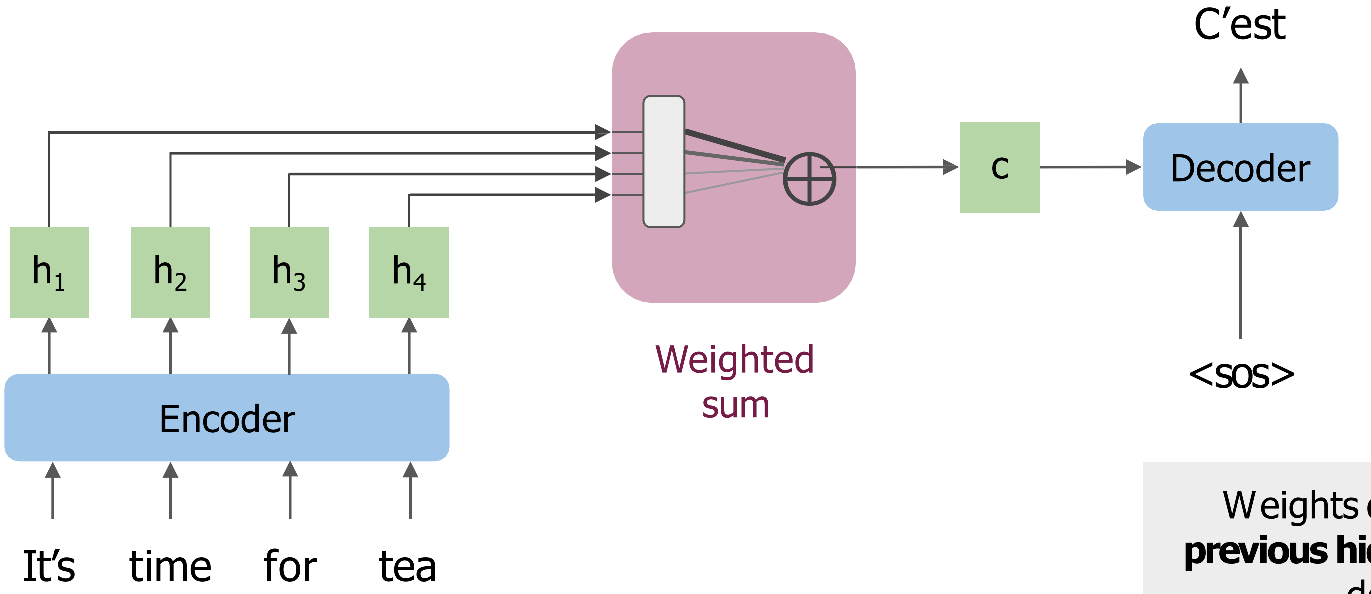
The model can focus on specific hidden states at every step



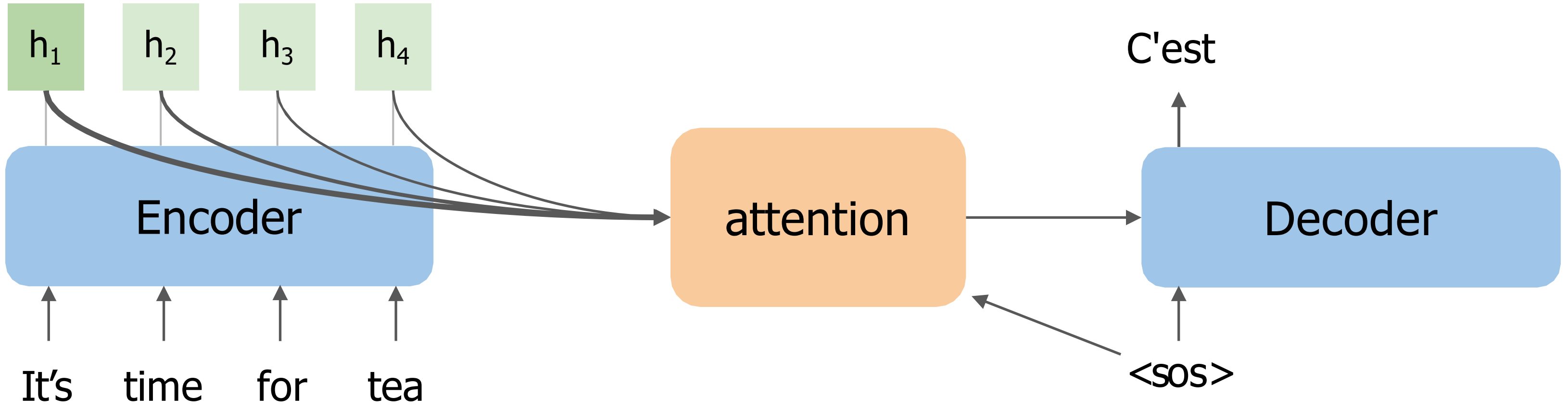
Suboptimal use of information



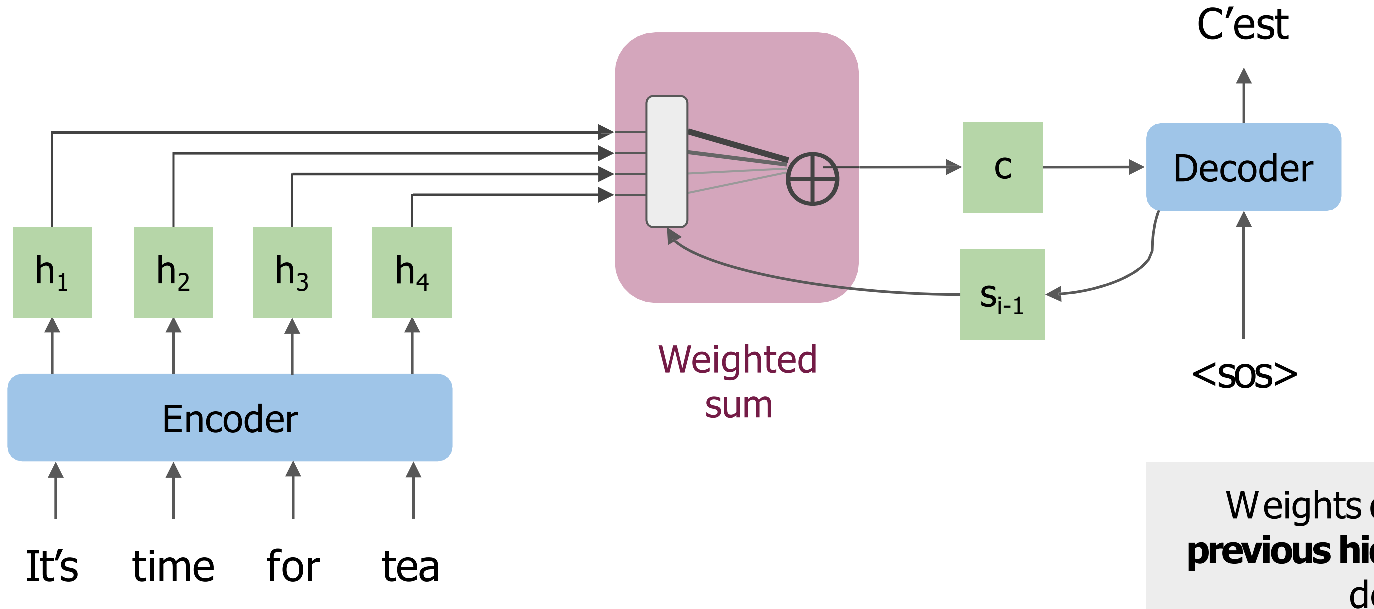
The model can focus on specific hidden states at every step



Weights depend on the **previous hidden state** in the decoder



The model can focus on specific hidden states at every step



Weights depend on the **previous hidden state** in the decoder

5. COMO FUNCIONAN LOS MODELOS SEQ2SEA CON MECANISMOS DE ATENCIÓN

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$$e_{ij} = a(s_{i-1}, h_j)$$

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k=1}^{T_x} \exp(e_{ik})}$$

h_j, s_{i-1} → Feedforward Network → e_{ij} → Softmax → α_{ij}

Learnable parameters

Weights used for the sum of hidden states

$$c_i = \sum_{j=1}^{T_x} \boxed{\alpha_{ij}}\boxed{h_j}$$

$$\alpha_{i1}h_1 + \alpha_{i2}h_2 + \alpha_{i3}h_3 + \cdots + \alpha_{iM}h_M \longrightarrow c_i$$

Context Vector is an expected value

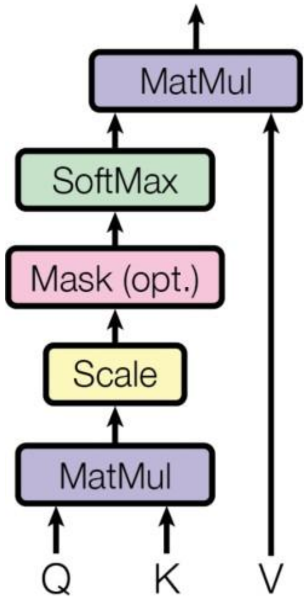
- Queries, Keys, and Values
- Alignment

Query

l’heure

Key	Value
It’s	[0.5, 0.2, -1.2, …,]
time	[0.2, -0.7, 0.9, …,]
for	[1.3, 0.3, 0.8, …,]
tea	[-0.4, 0.6, -1.1, …,]

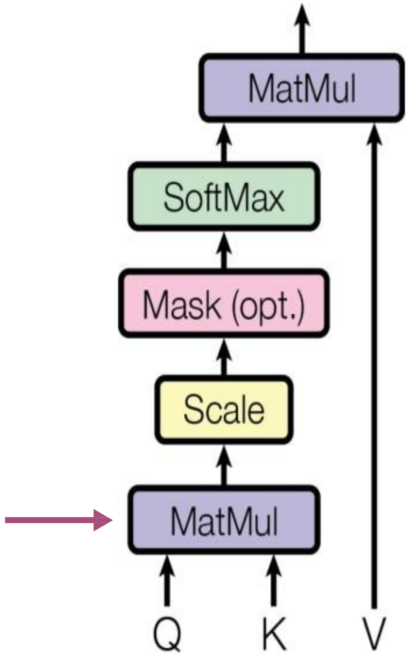
Similarity is used in for weighted sum



$$\mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

Queries Keys Values

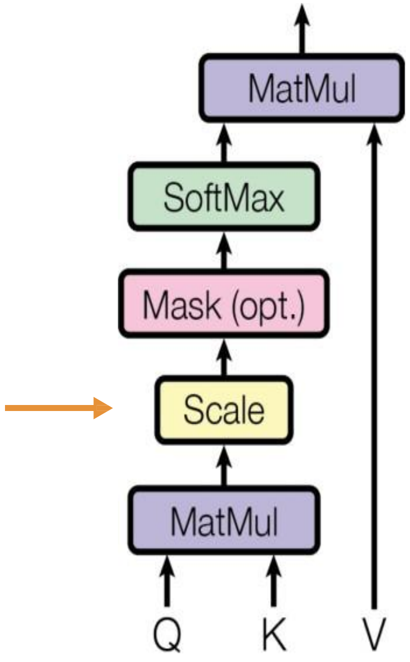
(Vaswani et al., 2017)



(Vaswani et al., 2017)

Similarity Between Q and K

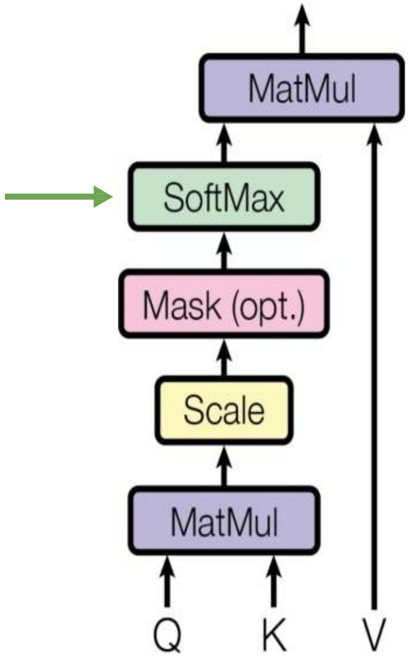
$$\mathrm{softmax}\left(\frac{\boxed{QK^\top}}{\sqrt{d_k}}\right)V$$



(Vaswani et al., 2017)

$$\mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

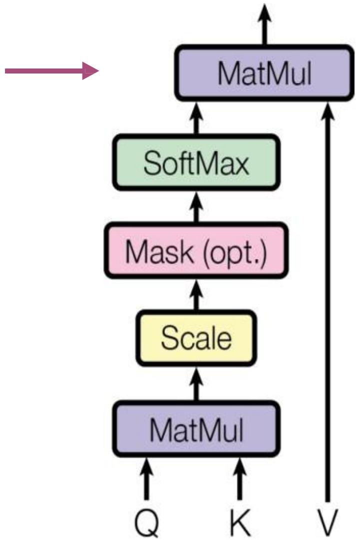
Scale using the root of the key vector size



(Vaswani et al., 2017)

$$\mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

Weights for the weighted sum



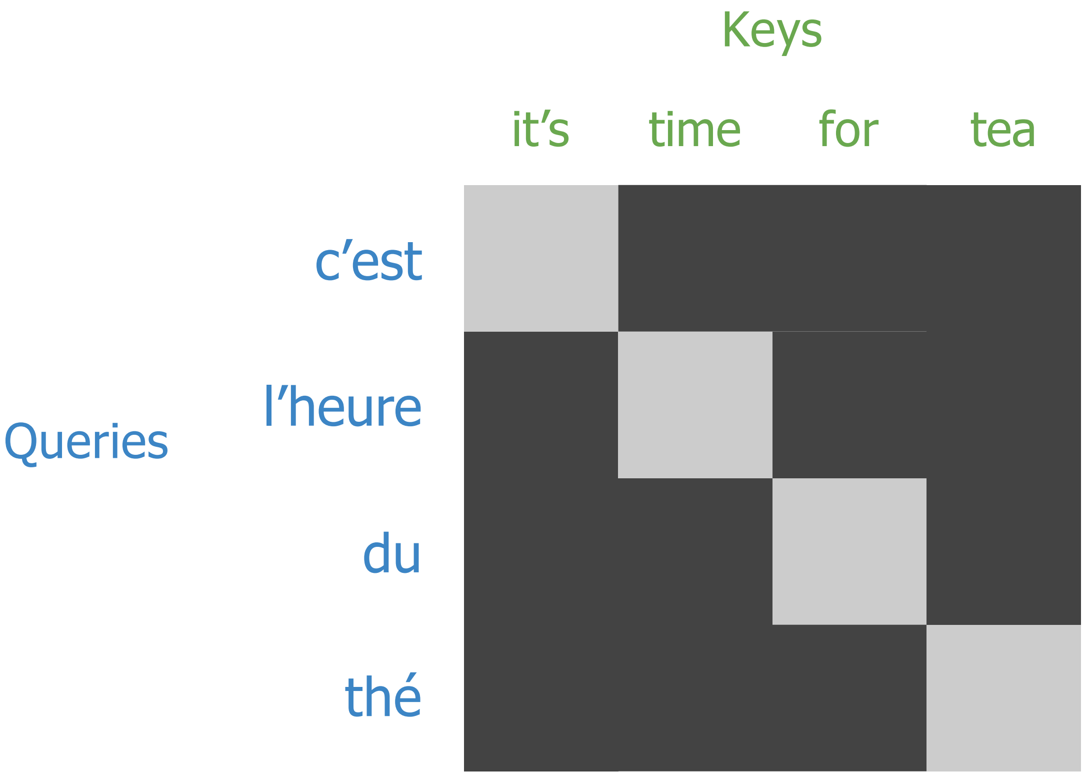
(Vaswani et al., 2017)

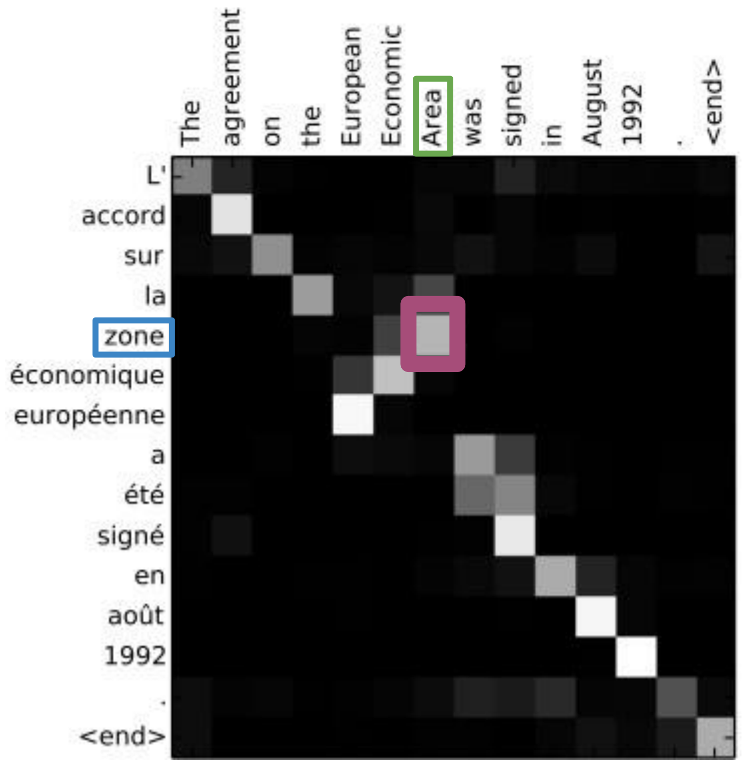
$$\mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

Weighted sum of values V

Just two matrix multiplications and a Softmax!

7. EJEMPLO DEL ALINEAMIENTO

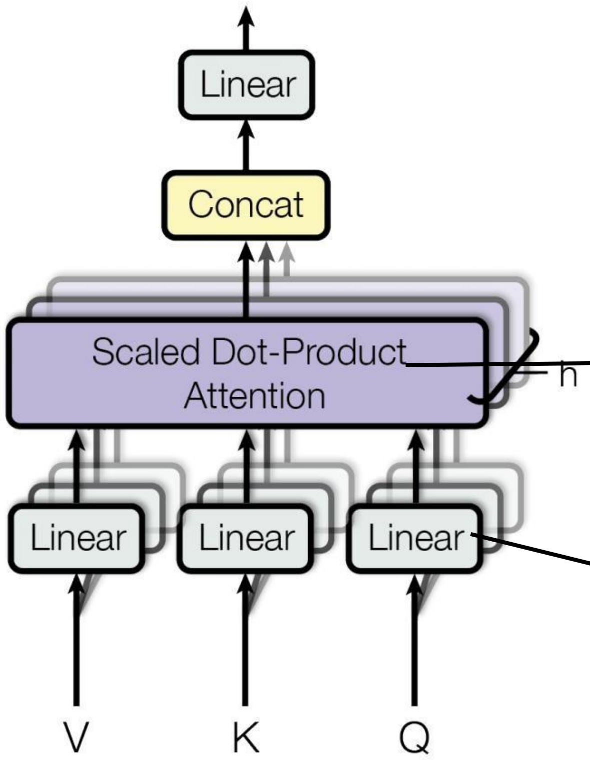




[Bahdanau et al., 2015]

Similar words have large weights

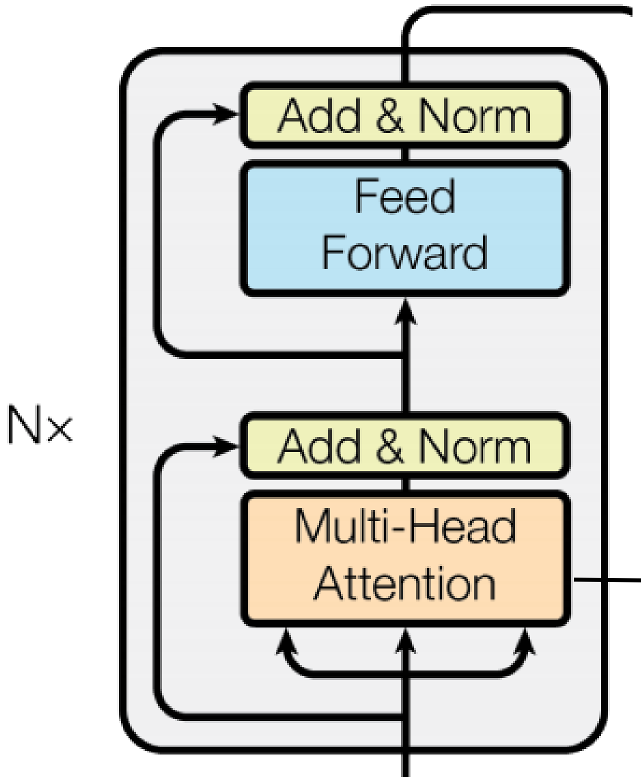
3. MULTI-HEAD ATTENTION



Scaled **dot-product attention multiple times** in parallel

Linear transformations of the input queries, keys and values

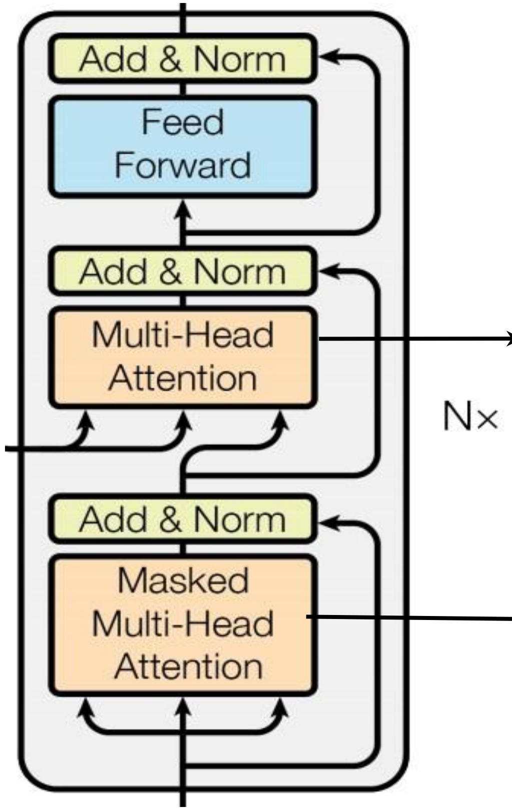
4. THE DECORDER



Provides contextual representation of each item in the input sequence

Self-Attention

Every item in the input attends to every other item in the sequence



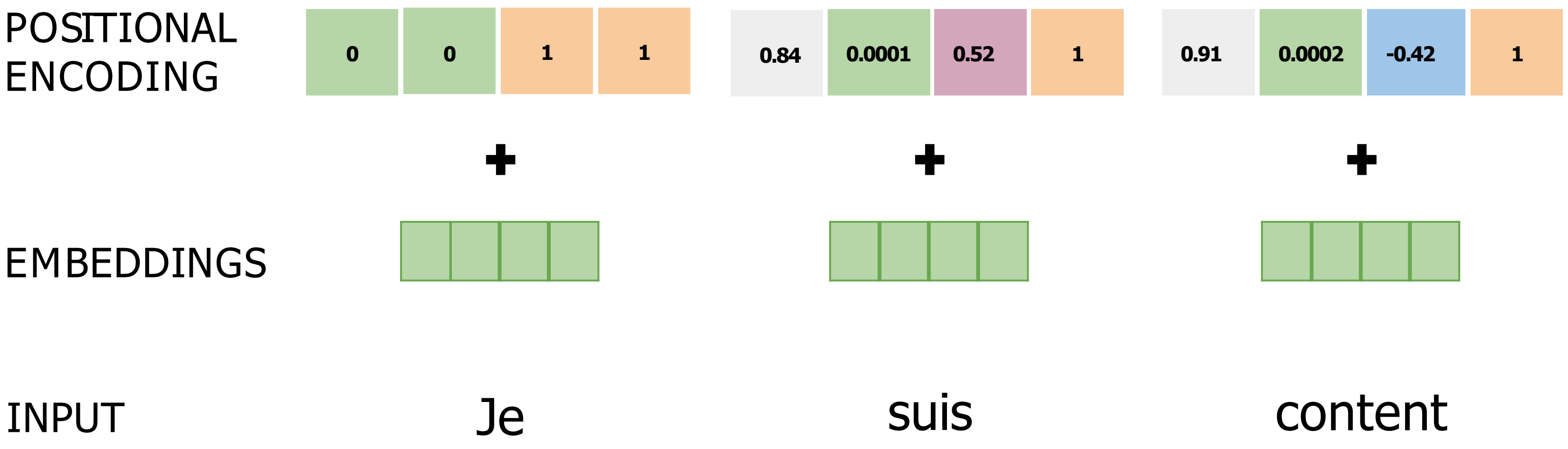
Encoder-Decoder Attention

Every position from the decoder attents to the outputs from the encoder

Masked Self-Attention

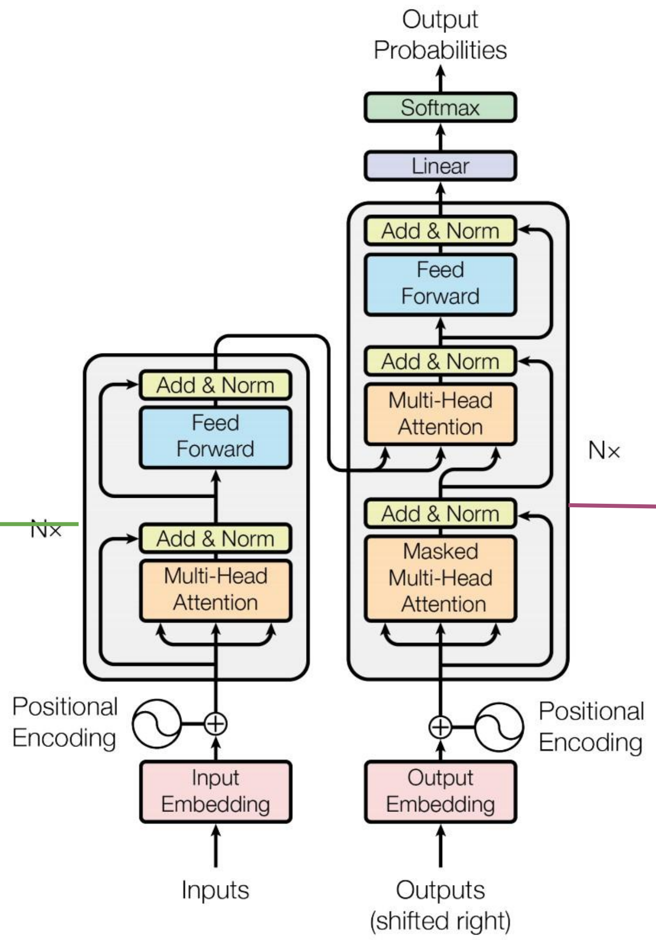
Every position attends to **previous** positions

6. POSITIONAL ENCODING



7. TRANSFORMER

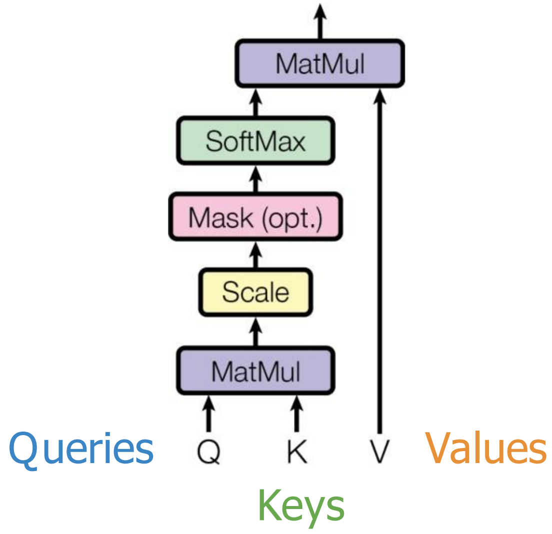
Encoder



Decoder

Easy to parallelize!

8. SCALED DOT PRODUCT DETAIL



(Vaswani et al., 2017)

Improves performance

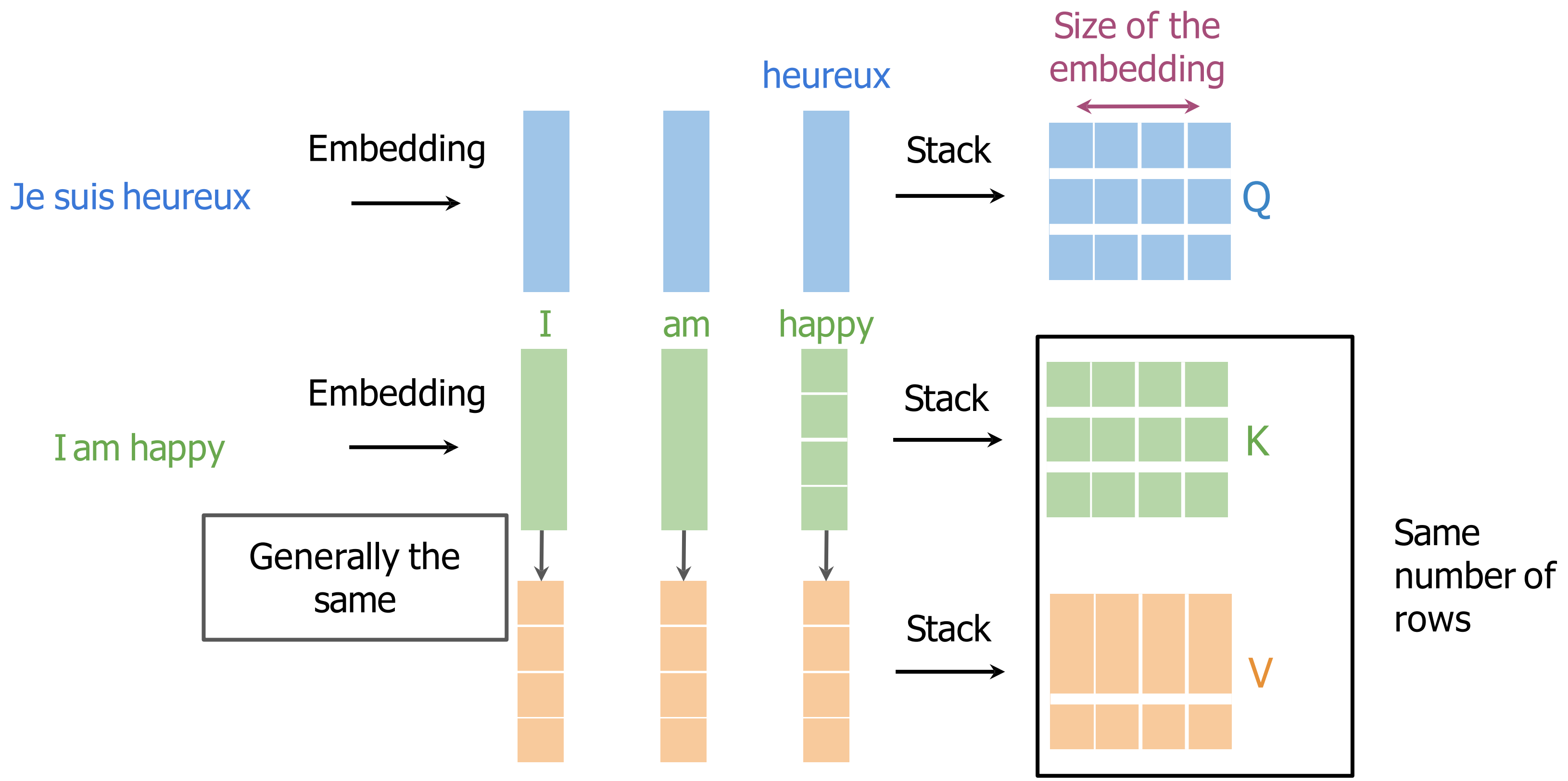
Weights add up to 1

$$\mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

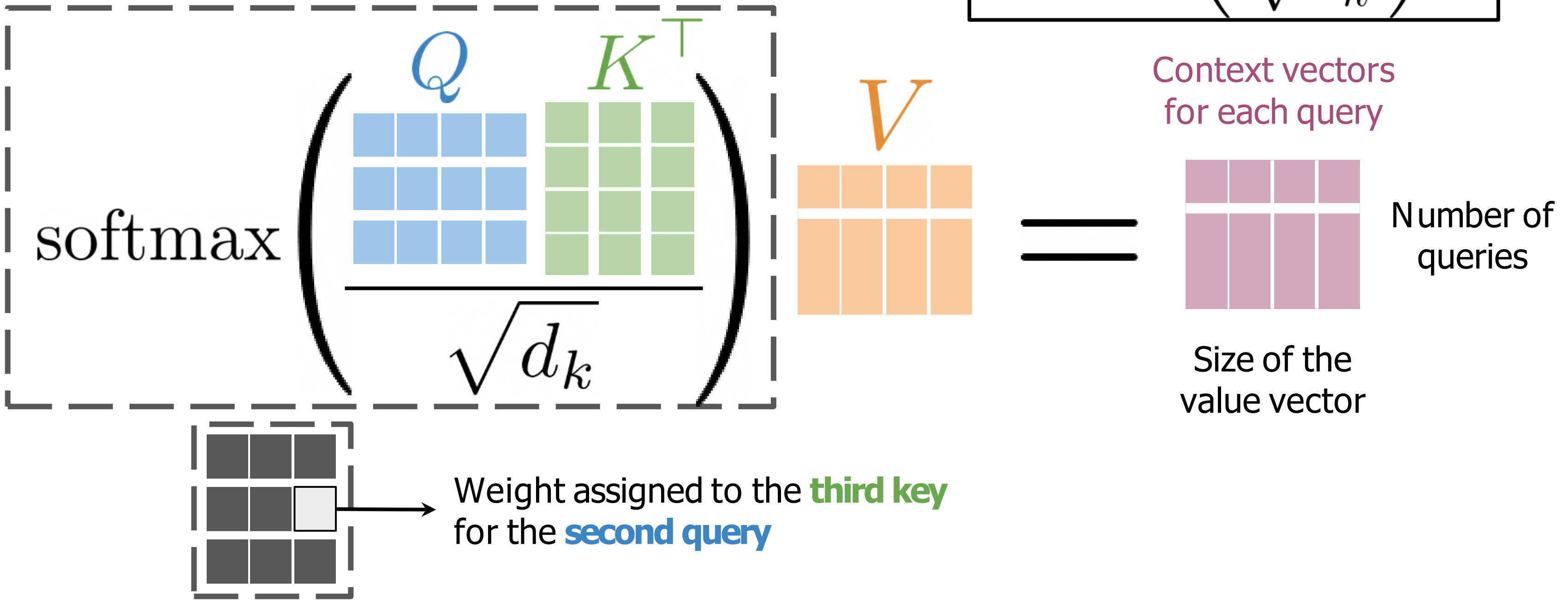
Weighted sum of values V

Just two matrix multiplications and a Softmax!

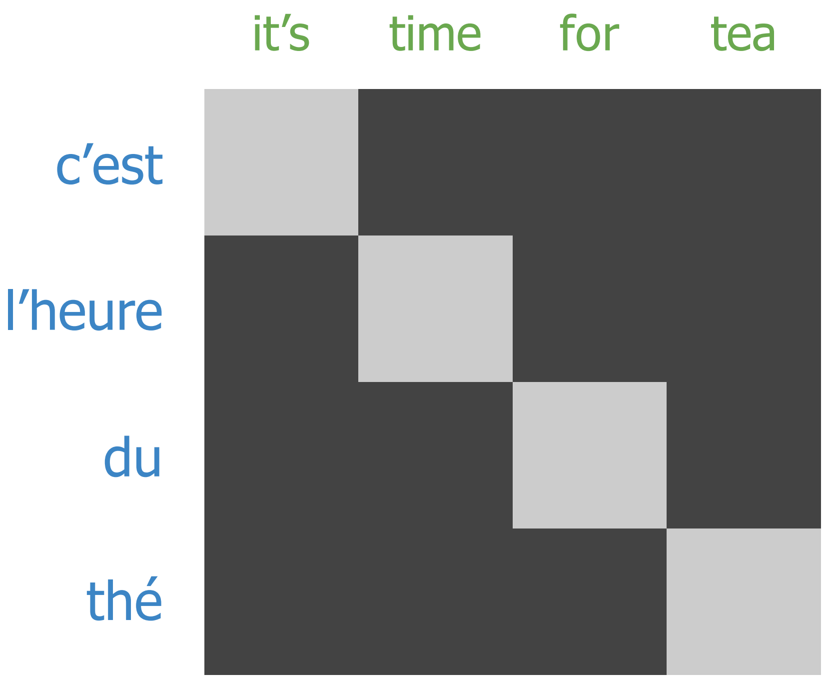
8. SCALED DOT PRODUCT DETAIL



$$\mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$



Queries from one sentence, keys and values from another



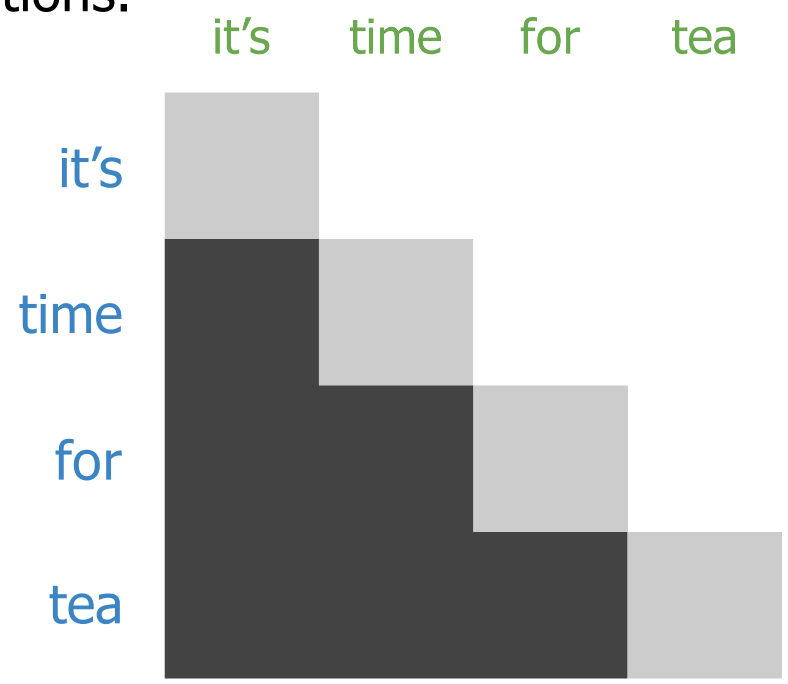
Weight matrix

Queries, keys and values come from the **same sentence**

Weight matrix

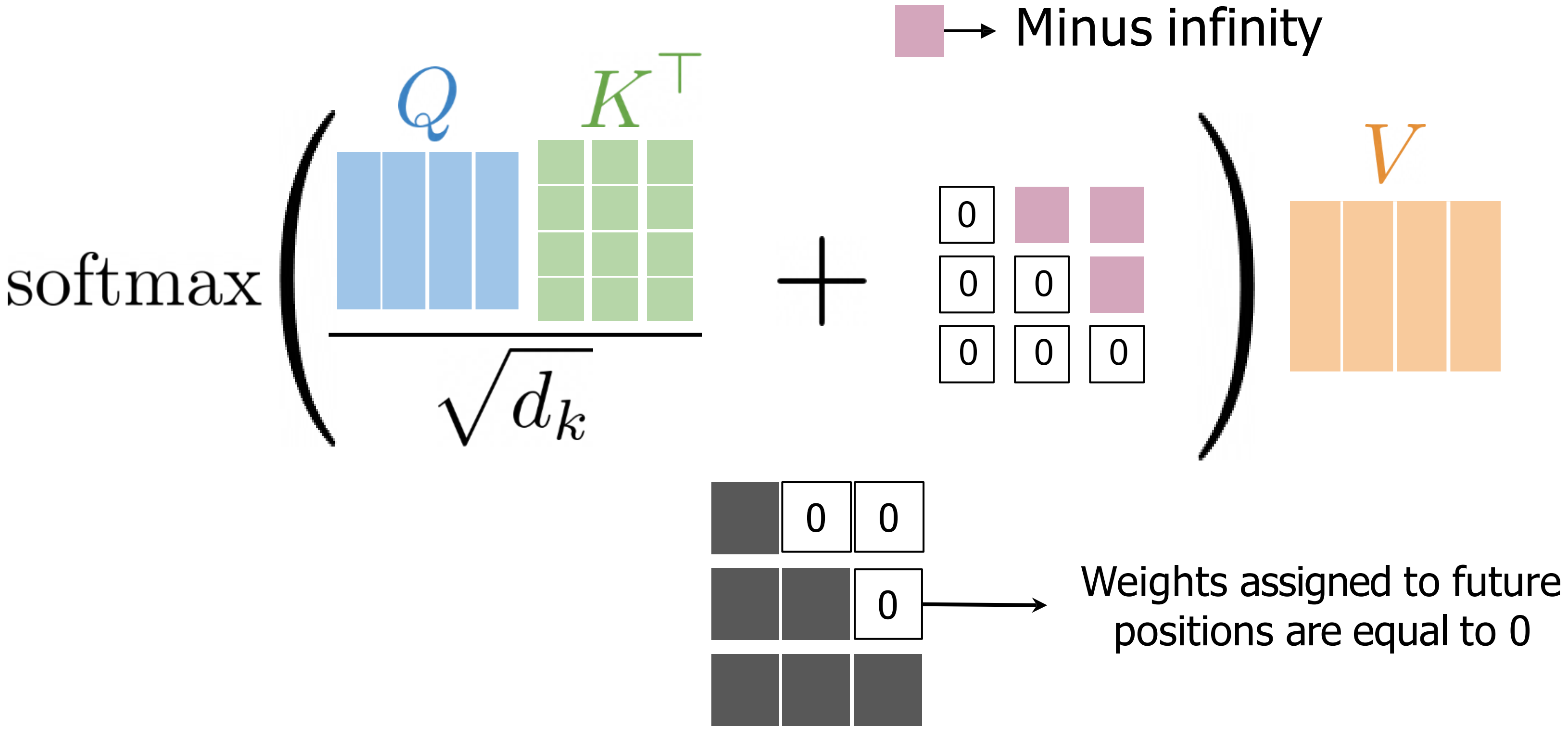
Meaning of each word **within** the sentence

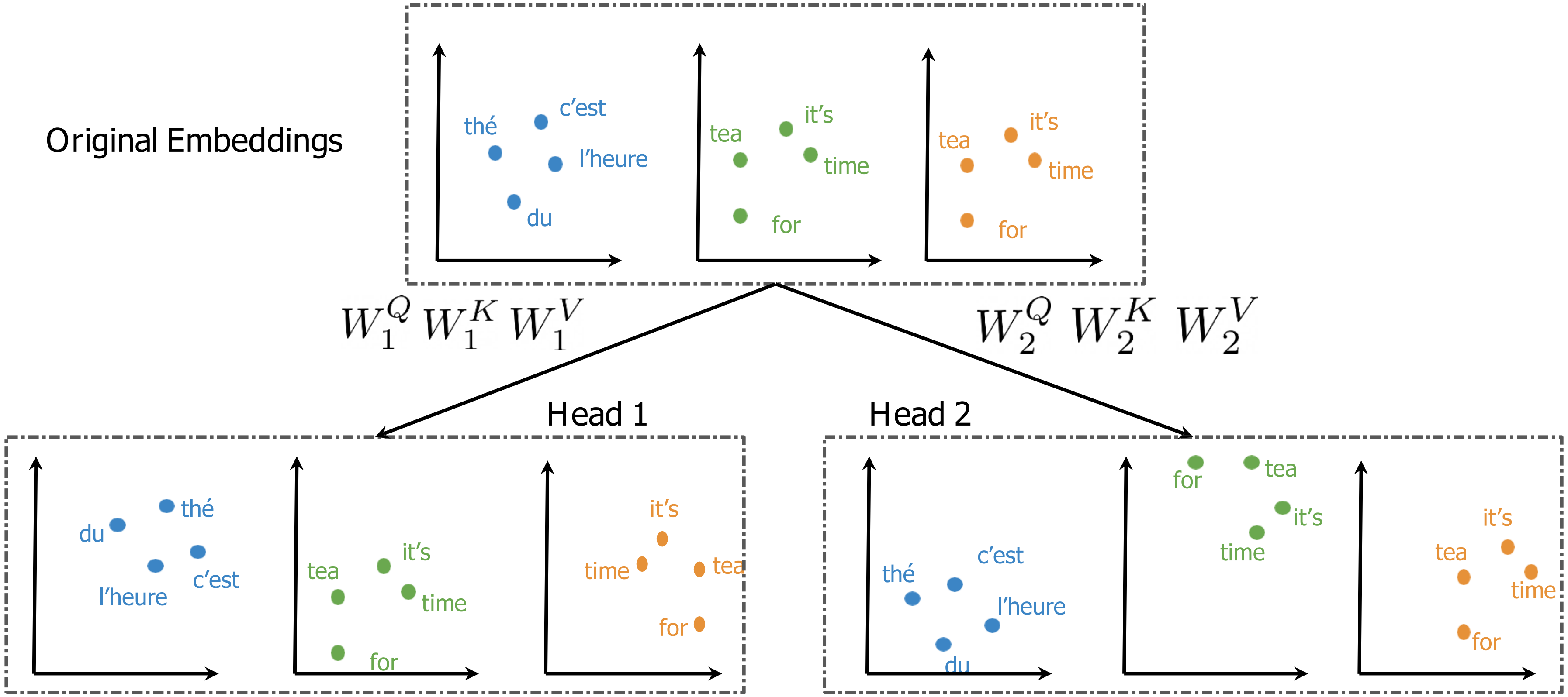
Queries, keys and values come from the **same sentence**. Queries don't attend to future positions.



Weight matrix

11. MASKED SELF-ATTENTION




Original Embeddings
thé
c'est
l'heure
du
tea
it's
time
for
tea
it's
time
for
$W_1^Q \; W_1^K \; W_1^V$
$W_2^Q \; W_2^K \; W_2^V$
Head 1
du
thé
c'est
l'heure
tea
it's
time
for
time
it's
tea
for
Head 2
thé
c'est
l'heure
du
for
tea
it's
time
tea
it's
time
for

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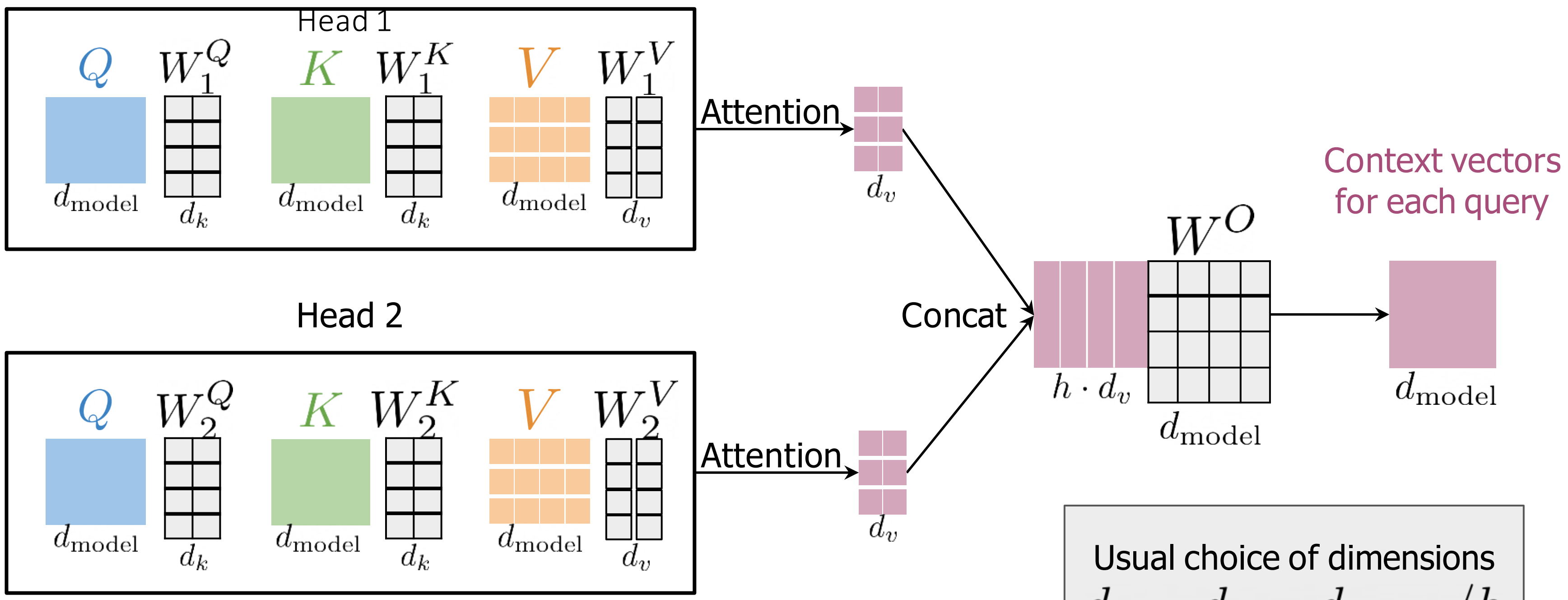


d_{model} : Embedding size

Usual choice of dimensions

$$d_k = d_v = d_{\text{model}}/h$$

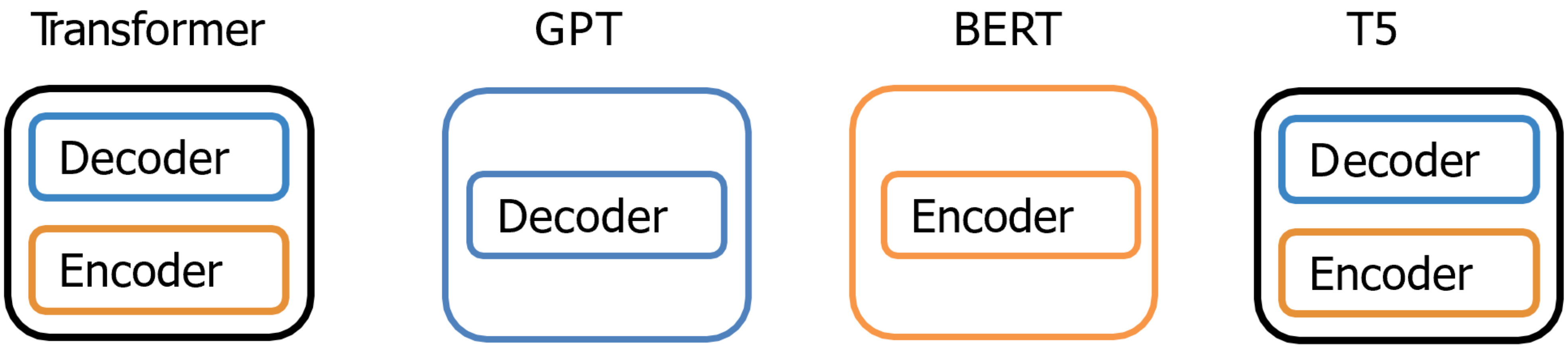
... they were $\underbrace{\text{on the}}_{\text{Fixed window}}$ right $\underbrace{\text{side of}}_{\text{Fixed window}}$ $\boxed{\text{the street.}}$

... they were on the right side of $\boxed{\text{history.}}$

The legislators believed that they were on the right side of history, so they changed the law.


2. ELMo, GPT, BERT, T5

CBOW	ELMo	GPT	BERT	T5
Context window	Full sentence	Transformer: Decoder	Transformer: Encoder	Transformer: Encoder - Decoder
FFNN	Bi-directional Context	Uni-directional Context	Bi-directional Context	Bi-directional Context
	RNN		Multi-Mask	Multi-Task
			Next Sentence Prediction	


Transformer
Decoder
Encoder
GPT
Decoder
BERT
Encoder
T5
Decoder
Encoder

BERT objective

Objective 1:
Multi-Mask LM

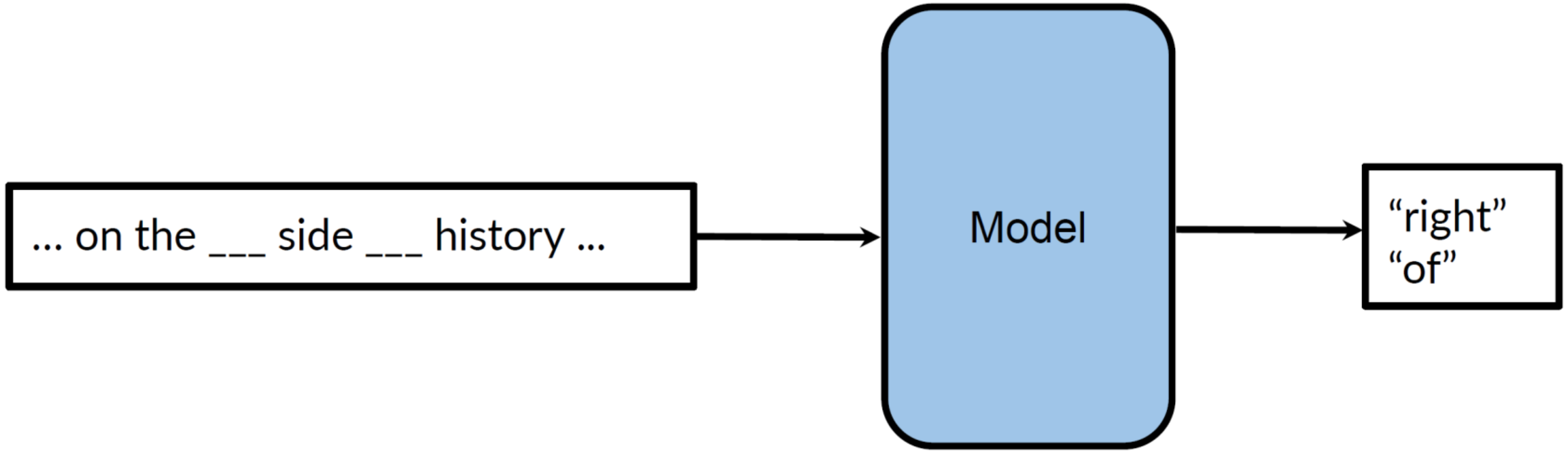
Loss: Cross Entropy Loss

Objective 2:
Next Sentence Prediction

Loss: Binary Loss

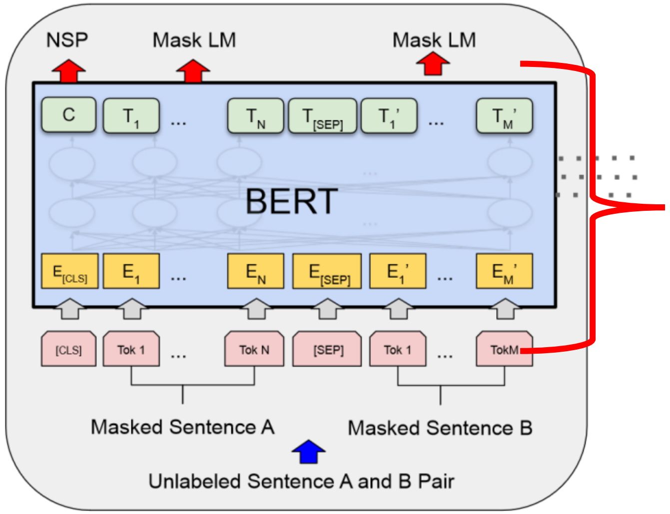
- After school ___________ his homework in the _________.
- Masked language modeling (MLM)
- After school Pajaro does his homework in the library.

Transformer + Bi-directional Context

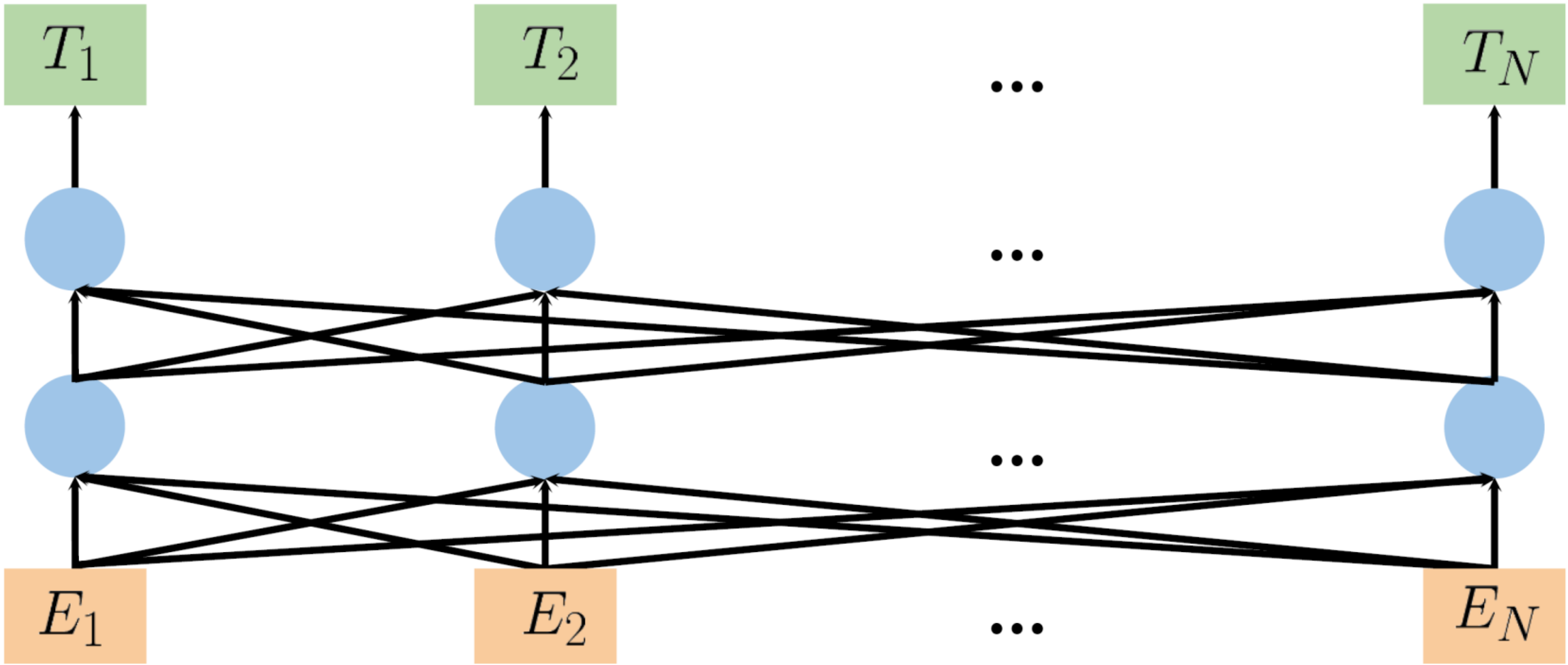


Multi-Mask Language Modeling

3. BERT: Pre-training



Pre-training



- A multi layer bidirectional “context” transformer
- Positional embeddings
- BERT_base:
  - 12 layers (12 transformer blocks)
  - 12 attentions heads
  - 110 million parameters

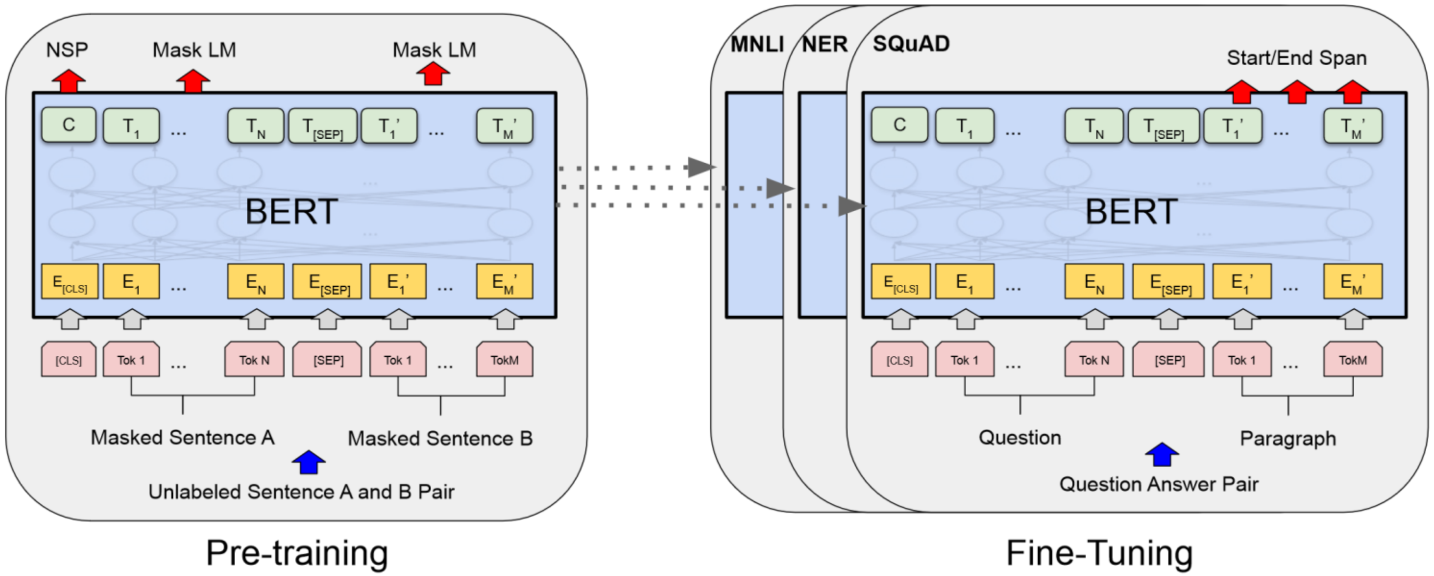


Figure 1: Overall pre-training and fine-tuning procedures for BERT. Apart from output layers, the same architectures are used in both pre-training and fine-tuning. The same pre-trained model parameters are used to initialize models for different down-stream tasks. During fine-tuning, all parameters are fine-tuned. [CLS] is a special symbol added in front of every input example, and [SEP] is a special separator token (e.g. separating questions/answers).

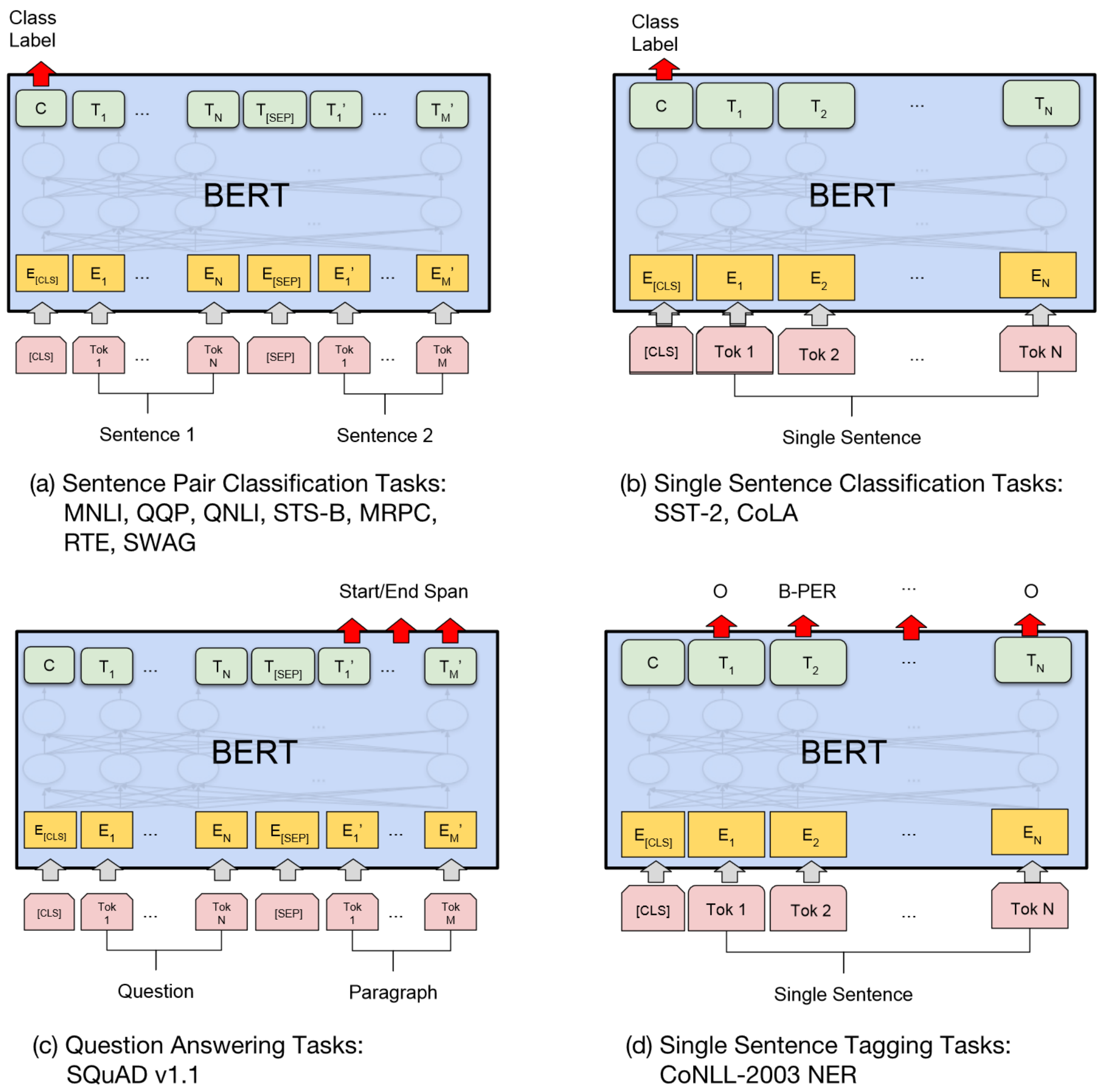
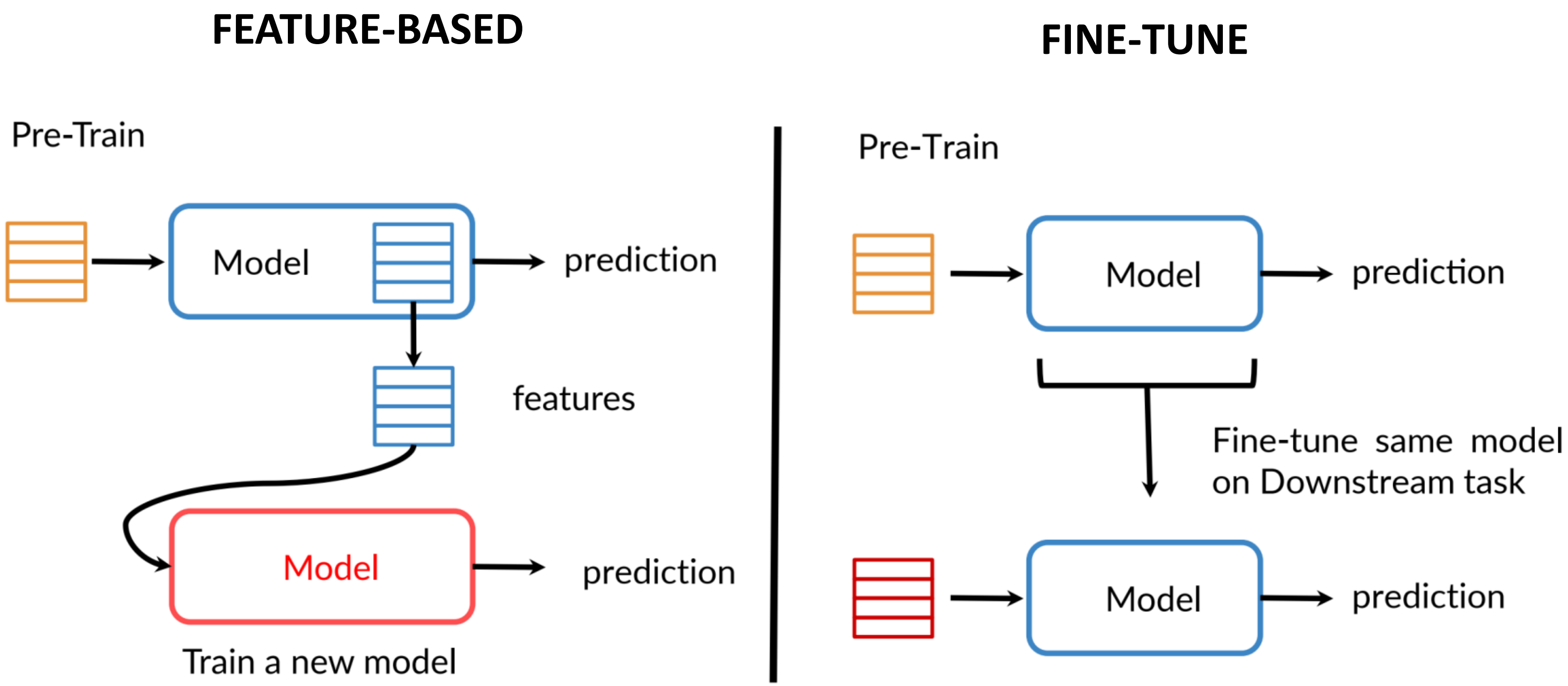
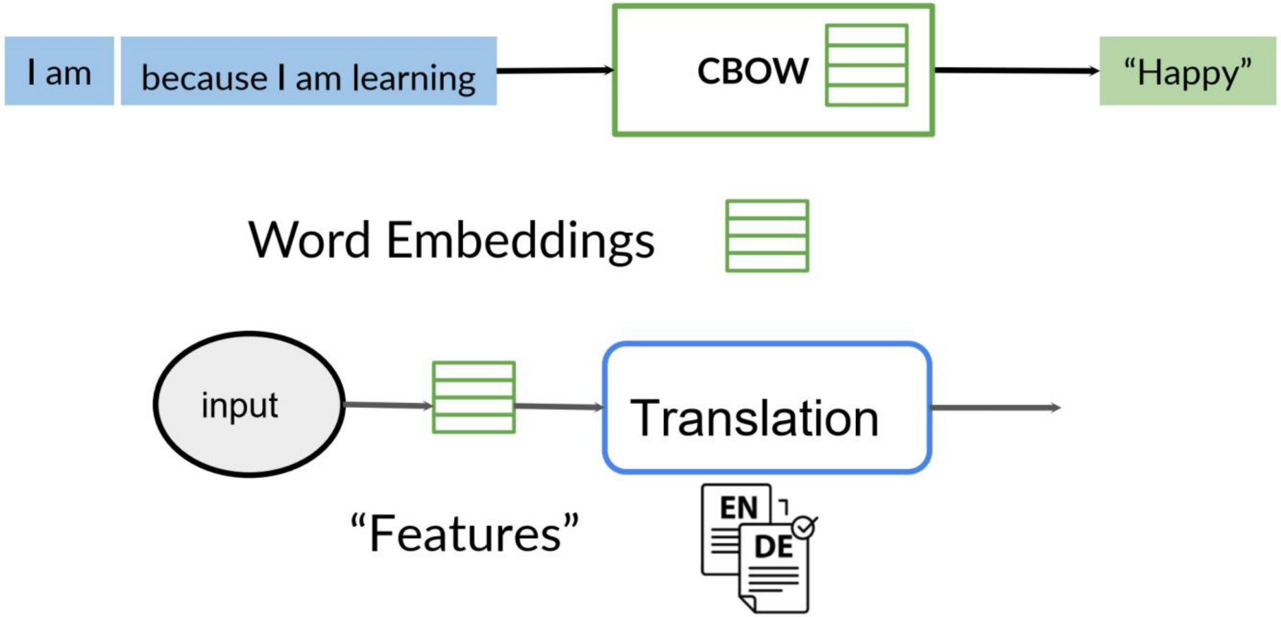
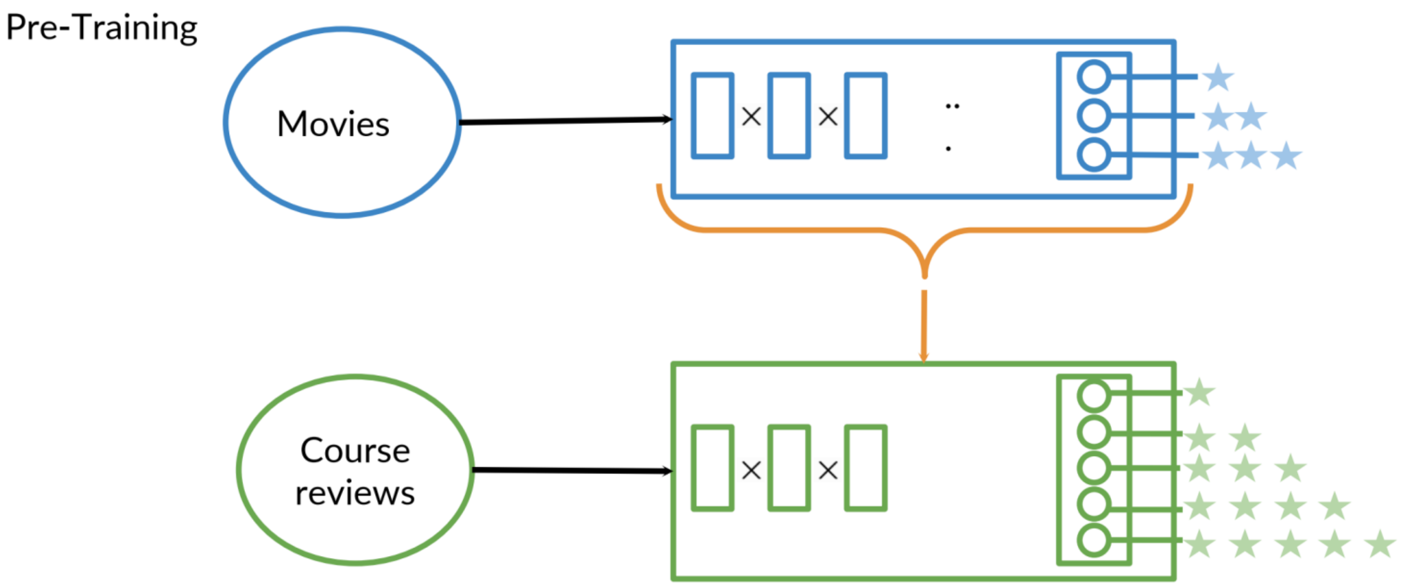


Figure 4: Illustrations of Fine-tuning BERT on Different Tasks.


FEATURE-BASED
FINE-TUNE
Pre-Train
Model
prediction
features
Model
prediction
Train a new model
Pre-Train
Model
prediction
Fine-tune same model
on Downstream task
Model
prediction


I am
because I am learning
CBOW
"Happy"
Word Embeddings
input
Translation
"Features"
EN
DE


Pre-Training
Movies
Course reviews